

SHAP (SHapley Additive exPlanations) - Structured Notes

1. Overview

SHAP is a method for explaining individual predictions of machine learning models by assigning each feature a contribution value.

Core idea:

■ A prediction is decomposed into a baseline plus contributions from each feature.

$$f(x) = \mathbb{E}[f(X)] + \sum_{i=1}^p \phi_i(x)$$

- $\phi_i(x)$: contribution of feature i for input x
 - $\mathbb{E}[f(X)]$: baseline (average model output under the chosen output scale)
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2. Local vs Global Importance

Local (primary purpose)

- SHAP explains **individual predictions**
- Each $\phi_i(x)$ is specific to a single observation

Global (derived)

- Aggregate local values across the dataset

Common metric:

$$\mathbb{E}[|\phi_i(X)|]$$

3. Formal Definition

Let:

- $N = \{1, \dots, p\}$: feature set
- $S \subseteq N$: subset of features

For a chosen value function $v_x(S)$ that measures the model output when only features in S are treated as known, the Shapley value is:

$$\phi_i(x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(p - |S| - 1)!}{p!} [v_x(S \cup \{i\}) - v_x(S)]$$

One common choice is the conditional value function:

$$v_x(S) = \mathbb{E}[f(X) \mid X_S = x_S]$$

4. Interpretation

$\phi_i(x)$ = average marginal contribution of feature i

- More precisely, compare the value function before and after adding feature i to a subset S
 - Average that marginal contribution across all possible subsets (contexts)
 - The meaning of "without feature i " depends on the chosen value function
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5. Role of Feature Order

Features have **no natural order**, but SHAP introduces order as a mathematical tool.

Key issue

Feature contribution depends on context due to interactions.

Solution

- Consider all possible orderings of features
- Measure the contribution when a feature enters
- Average over all orderings

This is equivalent to averaging over all subsets, with a weight determined by how many orderings produce each subset as the set before feature i .

6. Permutation View

$$\phi_i(x) = \mathbb{E}_{\pi} [v_x(\text{Pre}_{\pi}(i) \cup \{i\}) - v_x(\text{Pre}_{\pi}(i))]$$

- π : random permutation of features
 - $\text{Pre}_{\pi}(i)$: features before i in permutation π
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7. Where the Weighting Term Comes From

Weight assigned to subset S :

$$\frac{|S|!(p - |S| - 1)!}{p!}$$

Interpretation

$$\Pr(\text{Pre}_\pi(i) = S)$$

So the coefficient is the probability that S is exactly the set of features appearing before i in a random permutation.

Derivation

- Order the $|S|$ features in S before i : $|S|!$
 - Place feature i next
 - Order the remaining $p - |S| - 1$ features after i : $(p - |S| - 1)!$
 - Total permutations: $p!$
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8. Example (3 Features)

For feature A , let the other features be B and C .

All permutations and the set before A :

- $ABC \rightarrow S = \emptyset$
- $ACB \rightarrow S = \emptyset$
- $BAC \rightarrow S = \{B\}$
- $CAB \rightarrow S = \{C\}$
- $BCA \rightarrow S = \{B, C\}$
- $CBA \rightarrow S = \{B, C\}$

Resulting weights:

- $\emptyset$: $2/6 = 1/3$
- $\{B\}$ and $\{C\}$: $1/6$ each
- $\{B, C\}$: $2/6 = 1/3$

This matches the SHAP weighting formula exactly.

9. Why Weighting Matters

It ensures:

- Equal treatment of all permutations
- Fair representation of different interaction contexts
- Equal total weight for each subset size after summing over all subsets of that size

10. Linearity Property

For models f and g defined under the same value-function semantics:

$$\phi_i^{f+g}(x) = \phi_i^f(x) + \phi_i^g(x)$$

More generally:

$$\phi_i^{af+bg}(x) = a\phi_i^f(x) + b\phi_i^g(x)$$

Why it holds

- SHAP is defined via sums and expectations
 - Both are linear operations
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11. Implications of Linearity

(a) Linear models

For a linear model $f(x) = \beta_0 + \sum_i \beta_i x_i$, interventional SHAP gives:

$$\phi_i(x) = \beta_i(x_i - \mathbb{E}[X_i])$$

If features are dependent and you use conditional SHAP instead, attribution can be redistributed across correlated features.

(b) Additive models

If

$$f(x) = \sum_j f_j(x)$$

then

$$\phi_i(x) = \sum_j \phi_i^{f_j}(x)$$

(c) Tree ensembles

- Model = sum of trees
 - SHAP = sum of per-tree contributions
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12. Value Function Choices

Interventional SHAP

$$v_x(S) = \mathbb{E}_{X_S}[f(x_S, X_S)]$$

- Breaks dependence between observed and missing features
- Often easier to compute

Conditional SHAP

$$v_x(S) = \mathbb{E}[f(X) \mid X_S = x_S]$$

- Respects observed feature correlations
 - Harder to estimate and can redistribute credit across correlated features
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13. Key Properties (Axioms)

For a fixed value function, Shapley values are the **unique solution** satisfying:

1. Efficiency

$$\sum_i \phi_i(x) = v_x(N) - v_x(\emptyset)$$

With the usual SHAP baseline, this becomes:

$$\sum_i \phi_i(x) = f(x) - \mathbb{E}[f(X)]$$

2. **Symmetry** If two features contribute identically in all coalitions, they receive equal contributions.
 3. **Dummy** If adding a feature never changes the value function, its contribution is zero.
 4. **Linearity** Attributions add across summed models.
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14. Computational Considerations

Exact computation requires 2^p subsets per explained point.

Approximations:

- Kernel SHAP (sampling + regression)
 - Tree SHAP (efficient exact method for many tree models)
 - Deep SHAP (approximate method for neural networks)
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15. Final Summary

- SHAP is a **local explanation method** based on Shapley values
- It assigns contributions by averaging over all feature subsets
- The weighting term comes from counting permutations
- It ensures fair attribution under interactions
- Global importance is derived by aggregating local values

One-line takeaway:

SHAP values are the expected marginal contributions of features across all possible contexts, computed in a way that guarantees fairness and consistency for a chosen value function.